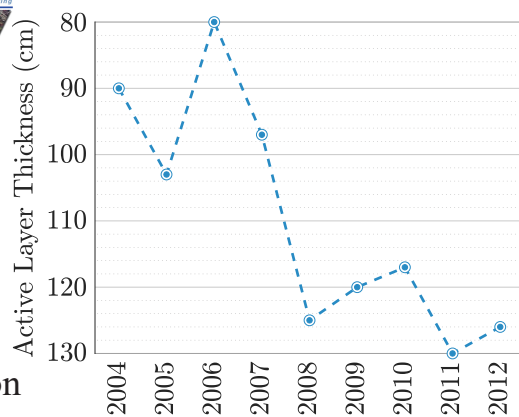


DATA CURATION AND PREPROCESSING

- Source: CALM datasets
- Measurement methods:
 - Mechanical probing (grids/transects)
 - Thaw tubes
 - Borehole temperature profiles
- Region: Circum-Arctic only ($\text{Lat} \geq 60^\circ\text{N}$)
- Period: 1962–2024
- Length of times series ≥ 8 years (129 sites)
- Gap filling (≤ 2 years) using PCHIP interpolation



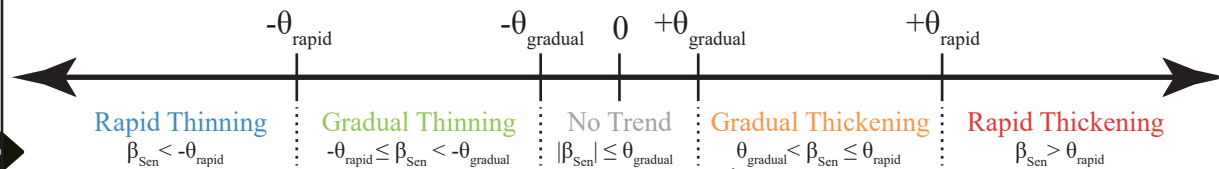
ACTIVE LAYER DYNAMICS INDEX CONSTRUCTION

Secular trend estimation

- Trend slope estimation
 - Sen's slope
 - Noise components
 - Min meaningful slope

Category assignment

- Threshold calibration
 - Rapid & gradual thresholds (from all sites)
- Reversal detection
 - Early and Recent years' slopes
- Slope-based classification



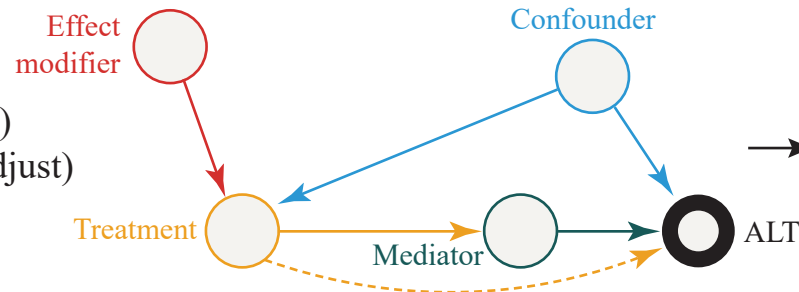
ALT CAUSAL INFERENCE



Data collection

DAG specifications

- Confounders (adjust)
- Mediators (do not adjust)
- Backdoor criterion



Panel estimation

- Fixed effects (inter-sites)
- First differences (intra-sites)

per ALDI category

Uncertainty & significance

- Uncertainty quantification
 - Block bootstrap
- Significance estimation
 - Mann–Kendall test (Hamed–Rao correction)
 - Bootstrap CI